universe fit within a Planck-sized nugget, yielding a density so great that it strains one's ability to find a fitting metaphor or an enlightening analogy: the density of the universe at the Planck time was simply colossal. At such energies and densities gravity and quantum mechanics can no longer be treated as two separate entities as they are in point-particle quantum field theory. Instead, the central message of this book is that at and beyond these enormous energies we must invoke string theory. In temporal terms, we encounter these energies and densities when we probe earlier than the Planck time of 10-41 seconds ATB, and hence this earliest epoch is the cosmological arena of string theory.

Let's head toward this era by first seeing what the standard cosmological theory tells us about the universe before a hundredth of a second ATB, but after the Planck time.

From the Planck Time to a Hundredth of a Second ATB

Recall from Chapter 7 (especially Figure 7.1) that the three nongravitational forces appear to merge together in the intensely hot environment of the early universe. Physicists' calculations of how the strengths of these forces vary with energy and temperature show that prior to about 10-35 seconds ATB, the strong, weak, and electromagnetic forces were all one "grand unified" or "super" force. In this state the universe was far more symmetric than it is today. Like the homogeneity that follows when a collection of disparate metals is heated to a smooth molten liquid, the significant differences between the forces as we now observe them were all erased by the extremes of energy and temperature encountered in the very early universe. But as time went by and the universe expanded and cooled, the formalism of quantum field theory shows that this symmetry would have been sharply reduced through a number of rather abrupt steps, ultimately leading to the comparatively asymmetric form with which we are familiar.

It's not hard to understand the physics behind such reduction of symmetry, or symmetry breaking, as it is more precisely called. Picture a large container filled with water. The molecules of H2O are uniformly spread throughout the container and regardless of the angle from which you view it, the water looks the same. Now watch the container as you lower the temperature. At first not much happens. On